

FLAb: Can Deep Learning Read an Antibody's Fitness Yet?

A Johns Hopkins benchmark of six protein models against physics-based Rosetta, across 13,384 antibody fitness measurements — and a sobering answer

Developing a therapeutic antibody is slow and expensive. A single candidate often has to be expressed, purified, and assayed for stability and binding before anyone knows whether it is worth pursuing. So it is only natural to want deep learning to speed things up: score thousands of candidate antibodies with a protein language model, keep the most promising ones, and take only those to the bench. But that shortcut works only if one assumption holds — that the model can actually tell a good antibody from a bad one.

That assumption has rarely been tested head-on. Antibody design methods are still commonly judged by weak proxies such as native sequence recovery, which say little about real therapeutic potential. And the major protein fitness benchmarks — CAFA, TAPE, FLIP — either exclude antibody data entirely or include only a sliver of it. The result is that no one has a principled way to answer a basic question: do these models understand antibody fitness at all, where fitness means how good a sequence is as a therapeutic across the properties that matter?

This paper fills that gap with FLAb — the Fitness Landscape for Antibodies — the largest therapeutic antibody design benchmark assembled to date. It gathers real experimental measurements and uses them to stress-test six widely adopted protein models against physics-based Rosetta. Before getting to the verdict, a pair of concrete examples makes the stakes tangible: on some datasets the model's ranking is sharp and trustworthy, while on others it collapses almost entirely.

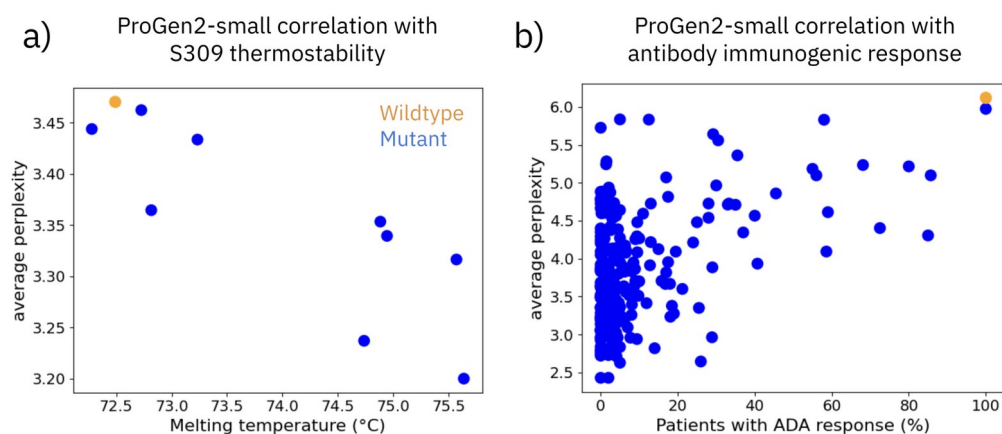


Figure 1. Two extremes of prediction quality. (a) On a thermostability dataset, the language model assigns lower perplexity — higher confidence — to the more thermostable, higher-melting-temperature variants, ranking them almost correctly ($r = -0.84$). (b) On an immunogenicity dataset, the model hands out both high and low confidence to therapeutics that all produce a 0% anti-drug-antibody (% ADA) response, showing essentially no discriminative power ($r = 0.48$).

Paper and code: bioRxiv preprint at <https://www.biorxiv.org/content/10.1101/2024.01.13.575504v1> ; open data and code at <https://github.com/Graylab/FLAb> . The work is by Michael Chungyoun, Jeffrey Ruffolo, and Jeffrey Gray of Johns Hopkins University (NeurIPS 2023).

Why antibodies need a benchmark of their own

A clinic-ready therapeutic antibody has to satisfy many competing demands at once: it must express well, stay structurally stable, avoid triggering an immune response, bind its target tightly, resist aggregation, and steer clear of sticking to the wrong things. These properties frequently pull against each other — improving one can degrade another. If a model could reliably score candidates across all of these axes, it could replace a great deal of slow, costly wet-lab screening.

But precisely because antibody evaluation is this specialized, general protein benchmarks do not transfer. As antibody-specific deep learning methods keep appearing, the field needs one shared, antibody-centric benchmark to converge on before trusting any model's predictions. FLAb is built to be exactly that.

What FLAb is: a bigger exam

FLAb draws on 8 separate studies to compile 17 mutational landscapes from distinct antibody families, totaling 13,384 experimental fitness measurements. Together they cover the six developability properties that define a therapeutic antibody: expression, thermostability, immunogenicity, aggregation, polyreactivity, and binding affinity. Crucially, each property is grounded in its own real experimental unit rather than an abstract score — which is what lets FLAb test whether a model genuinely generalizes across the many facets of antibody fitness.

Table 1. The six developability properties in FLAb and their experimental units. Following the paper, thermostability, aggregation, and polyreactivity are intrinsic properties, driven mainly by the antibody itself; binding affinity, expression, and immunogenicity are extrinsic, also shaped by target biology and mechanism of action. This split turns out to matter a great deal for the results.

Property	Experimental unit	Type
Thermostability	Melting temperature Tm (°C)	Intrinsic
Aggregation	Wavelength shift	Intrinsic
Polyreactivity	Retention time (min)	Intrinsic
Binding affinity	Dissociation constant KD (nM)	Extrinsic
Expression	Titer (µg/mL)	Extrinsic
Immunogenicity	% anti-drug antibody (% ADA)	Extrinsic

How to score an antibody — and how to grade the score

The evaluation is conceptually simple. Each antibody variable-region sequence (or its structure) is fed to a model, which reports a perplexity averaged over every residue in the heavy and light chains. Perplexity measures how surprised the model is by a sequence, so a well-behaved model should be confident — low perplexity — about high-fitness antibodies. The authors then correlate these scores against measured experimental fitness using three

coefficients: Pearson's r for linear trends, Spearman's ρ for monotonic trends, and Kendall's τ for ordinal agreement.

The important constraint is that every model is used exactly as released, with no fine-tuning. The six deep learning models span three architecture families — decoder-only (the ProGen2 suite, IgLM, ProtGPT2), encoder-only (AntiBERTy), and inverse-folding (ESM-IF, ProteinMPNN) — and all are compared against physics-based Rosetta energy.

Perplexity is defined as $\text{PPL}(s) = \exp(-\frac{1}{N} \sum \log p(x_i | x_{<i}))$, the exponential of the sequence-averaged negative log-likelihood. Lower values mean the model finds the sequence more natural and is more confident about it.

The big picture: a heatmap that stings a little

The clearest way to see the whole study is to compress every model-by-dataset pairing into a single figure. In the heatmap below, each cell's color encodes the strength of the correlation for that pairing: deeper blue means the model ranks antibodies correctly and usefully, deeper red means it ranks them backwards, and white means essentially no correlation at all.

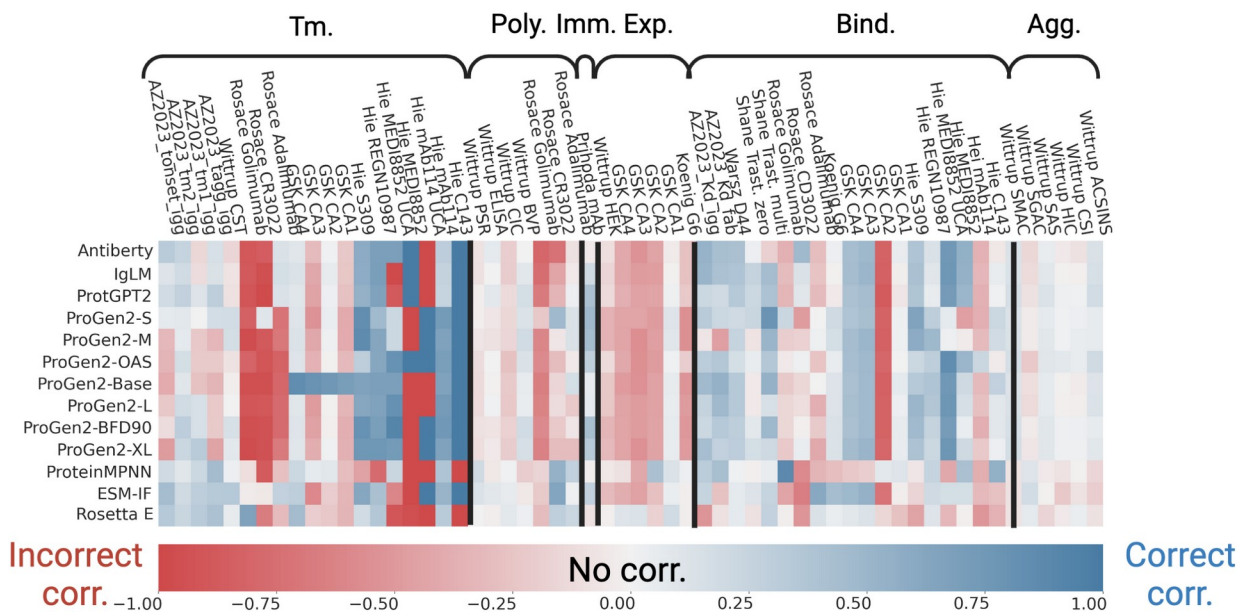


Figure 2. Performance for every model-dataset pair, as Pearson correlation coefficients (PCC). The sign was inverted for aggregation, expression, and thermostability so that a useful correlation always shows up as positive (blue). The large swaths of white and red are the point: on many datasets, the models do no better than chance, and sometimes point the wrong way.

The headline is a reality check: no model correlates well with all six properties, and even within a single property, performance swings sharply from one dataset to the next. Counting outright wins, ProGen2-Small tops the most datasets, coming first on 7, with ProGen2-Medium, ProGen2-OAS, ESM-IF, and Rosetta energy tied just behind at 6 each. The winner's circle, in other words, is crowded — nobody leads across the board.

Taking it apart: what actually drives performance

One consistent trend is worth flagging: on average, sequence-based models outperform structure-based models across all six properties, with the largest gap on thermostability. This suggests that, in the current setup, raw sequence signal still carries most of the predictive power.

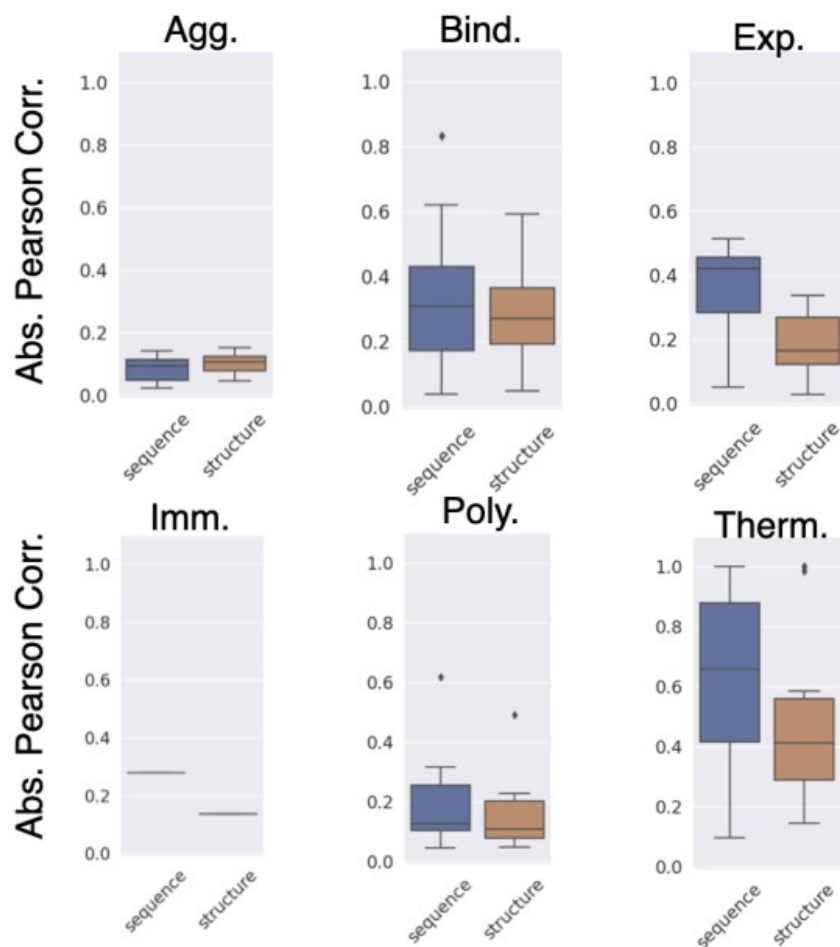


Figure 3. Sequence-based versus structure-based models, as absolute Pearson correlation across the six properties. The sequence bars average AntiBERTy, IgLM, ProtGPT2, and ProGen2; the structure bars average ProteinMPNN, ESM-IF, and Rosetta. Sequence models hold their own on every property and pull clearly ahead on thermostability (Therm.).

So why is the same property sometimes easy and sometimes impossible? A key factor is family. When the antibodies being compared come from the same family — often differing by only a few point mutations — the ranking is fairly reliable. But asking the model to compare antibodies across different families, of distinct origins, and the ability nearly vanishes.

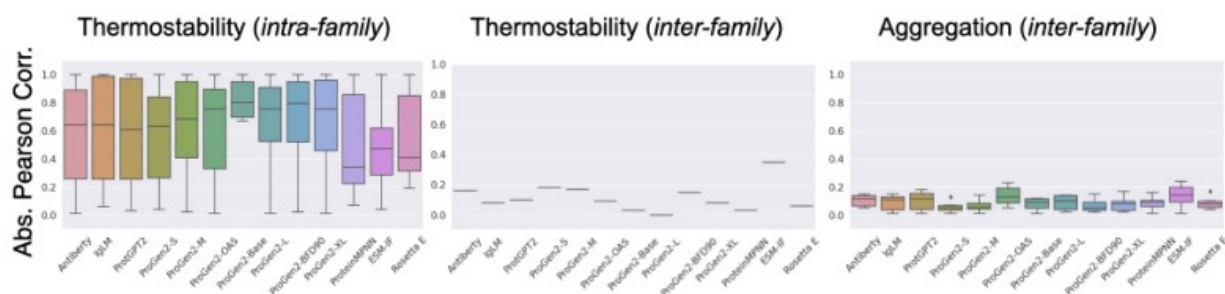


Figure 4. Intra-family versus inter-family comparisons. Correlations for intra-family thermostability (left) are consistently high; inter-family thermostability (middle) hugs zero; and inter-family aggregation (right) is just as weak. Models are far better at distinguishing variants within one family than at judging across families.

The gap is stark in the numbers: intra-family thermostability reaches an average absolute correlation of about 0.77, while inter-family sits at roughly 0.12. The same pattern shows up along the intrinsic-versus-extrinsic axis. Intrinsic properties (thermostability, aggregation, polyreactivity) average above 0.6 in absolute correlation, whereas the extrinsic ones lag well behind — binding affinity under 0.4, expression under 0.42, and immunogenicity under 0.5. Models read the properties driven by the antibody itself far better than those that depend on target biology.

The authors also dissect two factors that often draw attention. The first is architecture versus training data: encoder-only AntiBERTy and decoder-only IgLM, both trained on the same 558M OAS antibody sequences, perform almost identically, so architecture alone makes little difference. The second is parameter count: scaling ProGen2 from 151M up to 6.4B parameters noticeably improved only two properties, polyreactivity and thermostability. Parameter size matters more than architecture or training-data composition, but it is no cure-all either.

A final finding carries a warning. Language models can be biased toward evolutionary signal. For the clinically approved antibody golimumab, several language models rank the wild-type as fitter than mutants that are in fact more thermostable — while physics-based Rosetta correctly places the stabilized mutants ahead.

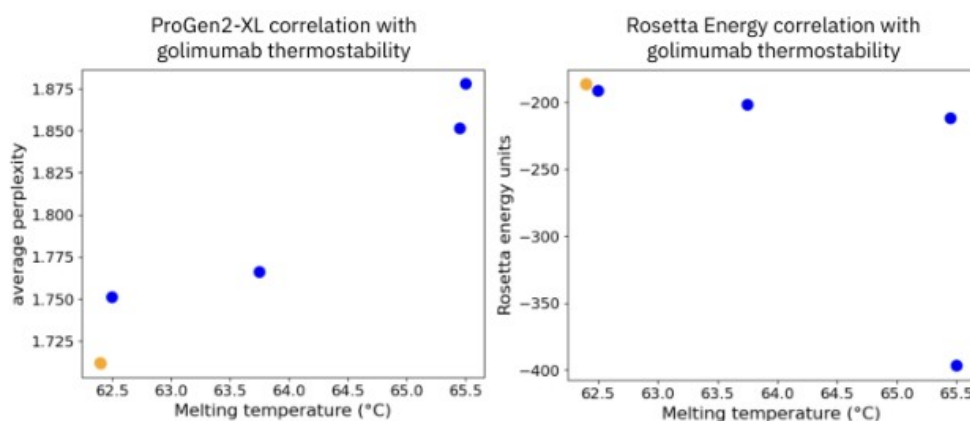


Figure 5. One case of evolutionary bias (golimumab mutants). Left: ProGen2-XL assigns the lowest perplexity — its highest confidence — to the wild-type (orange, lower melting temperature), a ranking skewed by evolutionary likelihood. Right: physics-based Rosetta correctly gives the more thermostable variants better stability scores. Evolutionarily common is not the same as physically better.

What this work tells us

FLAb's answer is honest and useful: as of today, no deep learning model reliably predicts antibody fitness across all developability properties. We are therefore not yet at the point of trusting these models to filter therapeutic candidates on their own. The encouraging part is that intrinsic biophysical properties are already captured reasonably well, which is precisely what points to where the hardest remaining work lies.

The authors argue that the most promising path forward is to give these models the information they currently lack — structure, antigen context, and physics-based priors — while continuing to grow open antibody fitness datasets like FLAb. All 13,384 measurements are released openly so the community can keep making this exam larger and more complete. For anyone hoping to accelerate antibody design with a model, FLAb draws a clear line: validate on real experimental data first, then talk about trust.